INTEGRATIVE MODELING OF DOXORUBICIN RESPONSE IN CANCER

Department of Genetics and Bioengineering

BIOE 492: Senior Design Project

Supervised by Asst. Prof. Dr. Özlem Ulucan Açan

June 16, 2026

Cansu Akdemir 122201043

Efehan Evren 121201060

Esra Rençberler 122201045

Istanbul Bilgi University

Faculty of Engineering and Natural Sciences

Table of Contents

List of Figures

List of Tables

Abstract

For the current study, the CCLE gene expression data and PRISM IC50 data were used to predict the sensitivity of lung cancer cell lines to doxorubicin. The lung cancer cell lines having IC50 values of doxorubicin were identified in the CCLE database, and only lung cancer cell lines were considered in this dataset. There were 96 lung cancer cell lines in the current dataset, and the IC50 values were transformed into log10(IC50). 457 genes were selected by feature selection technique based on Pearson correlation, and Random Forest was used as the baseline model for predicting doxorubicin sensitivity.

As shown in the result of 5-fold cross validation, the 60-gene signature was chosen as the final predictive signature since it resulted in the highest overall R^2. Functional enrichment analysis suggested that the final gene signature was mainly associated with cytosolic, vesicle-related, endosomal, and intracellular organization terms. The final gene signature includes GRN, NUP98, MAP2K2, KLF4, RBM39, EEF2, and WIPI2, which are genes involved in chemo-resistance or chemo-response pathways. Thus, the 60-gene transcriptomic signature may serve as a potential predictive model for doxorubicin sensitivity in lung cancer cells.

1. INTRODUCTION

Doxorubicin is a chemotherapy drug based on anthracyclines which have been widely used for cancer treatment for many years. The antitumor activity of doxorubicin is associated with its ability to cause DNA damage, inhibit topoisomerase II, generate reactive oxygen species, and activate apoptotic processes. As a result, the cytotoxic activity of doxorubicin becomes pronounced, especially for rapidly proliferating cancer cells. Nevertheless, cancer cell responses to chemotherapy can vary due to tumor heterogeneity [11]. While some cancer cells are sensitive to the effects of doxorubicin, other cells demonstrate less sensitivity to the drug.

The development of resistance towards drugs used for treating cancer is one of the key factors that affects the efficiency of cancer therapy. Doxorubicin resistance can be linked to various molecular processes such as drug efflux transporters, apoptosis, increased DNA repair, oxidative stress response, autophagy, survival signals, and heterogeneity. Hence, it may not always be sufficient to explain the response to doxorubicin on the basis of one particular gene or pathway alone. Evaluating the combined effect of changes at the gene expression level would help achieve this goal.

Lung cancer refers to the kind of cancer characterized by molecular heterogeneity, which causes considerable variation in the response to specific treatment options among individuals or cell lines [11]. This heterogeneity often poses challenges in the prediction of drug response due to its complex nature. Transcriptional profiling provides critical information that helps in understanding the expression profiles that indicate sensitivity or resistance to certain drugs, since the transcription profile represents the status of the cancer cells.

CCLE gene expression data used in this work offer transcriptomic data of cancer cell lines [1]. On the other hand, the PRISM drug response dataset has cell line IC50 values of various compounds [4]. Combining the two datasets using DepMap ID facilitated the examination of the association between doxorubicin response and the pattern of gene expression [1,4]. The IC50 values served as the continuous regression targets, which were further transformed into log10(IC50).

However, even though there are usually thousands of genes in such databases, the number of samples is fewer. Hence, high-dimensional problems may arise from this. Furthermore, the chances of overfitting may increase. Thus, feature selection becomes an essential phase in the process of drug response modeling [5]. For instance, in our work, the genes that were related to log10(IC50) were selected through feature selection. Afterward, a Random Forest regression model was generated, and gene subsets were created using RFE [5].

The primary goal of this research is to find out a compressed set of transcripts that could be used for the identification of doxorubicin sensitivity. In order to achieve this aim, first, the performance of baseline Random Forest was assessed. Then, the different gene sets selected by RFE were compared using 5-fold cross validation, and then the best gene set was chosen. At last, the functional enrichment analysis of the best chosen gene set was conducted.

2. METHODOLOGY

2.1. Overall Workflow

In this study, an integrated pharmacogenomic approach has been utilized to predict doxorubicin sensitivity. In particular, CCLE gene expression data as well as the PRISM drug sensitivity data sets were used [1,4]. The IC50 values of doxorubicin were then selected from the PRISM data set. Records with no IC50 values were deleted. Finally, the two data sets were aligned based on the shared DepMap ID.

The following filtering steps were performed on the integrated data for lung cancer cell lines. The IC50 values in the filtered data were converted into numeric values and then transformed by taking their log10(IC50). Other than the depmap_id, raw IC50, and log10(IC50), all other features present in the gene expression matrix were selected. All genes having zero variance were removed, and feature selection based on Pearson correlation was done.

In the modeling stage, a baseline random forest regression model was first built with the selection of correlated genes. The effectiveness of this model was assessed through 5-fold cross-validation with the measures of MAE, RMSE, and R^2 scores. Following this, recursive feature elimination was used to generate a varying number of genes. Each new group of genes was again validated using 5-fold cross-validation and ranked according to their Overall RMSE and R^2 scores. The final gene signature was selected based on the maximum Overall R^2 score obtained from RFE-generated groups of genes.

2.2. Dataset Integration: CCLE Expression and PRISM IC50

In this analysis, the gene expression dataset utilized was that of the CCLE while the drug response dataset used is that of the PRISM secondary screen dose-response [1,4]. The only records chosen were those where the value of the drug name column was "doxorubicin". All cell lines that did not have an IC50 for doxorubicin were removed from further analysis. This filtering led to the remaining 471 cell lines having an IC50 for doxorubicin.

In the expression dataset from CCLE, the first column that served as an identifier for the cell line was changed to depmap_id. Common DepMap IDs among the datasets of CCLE expression data and PRISM doxorubicin IC50 data were then found. There were a total of 460 common cell lines that had both gene expression data and doxorubicin IC50 values. Subsequently, the expression dataset was sorted using these common cell lines, and only the depmap_id and IC50 columns were taken from the PRISM dataset.

During the last step, the merging between expression data and IC50 data was performed by means of the depmap_id. The resulting combined dataset comprised information of rows as cell lines and columns as expression features, and IC50 doxorubicin values. Such a dataset was the starting point for further filtering and prediction modeling of lung cancer data.

2.3. Lung Cancer Cell Line Filtering

Following construction of the combined data set, the analyses were conducted exclusively on the lung cancer cell lines. The sample_info data set was used to select cell lines that had the keyword "lung" in the primary_disease category. This search was done without regard for case sensitivity so that all entries of lung cancer under primary_disease were selected.

The DepMap IDs for lung cancer that had been extracted from the sample_info dataset were matched to the expression-IC50 dataset. Only those cell lines that matched on both lists were used in the modeling process. Consequently, the modeling dataset comprised a total of 96 lung cancer cell lines. This way, the modeling process could be done for a particular cancer type.

It facilitated the construction of a predictive model for doxorubicin response based on gene expression in a disease subtype specific setting. The use of lung cancer cell lines would help to minimize biological variability across different diseases and to study the gene expression-drug response correlation in a relatively homogenous group.

2.4. IC50 Log10 Transformation

IC50 for doxorubicin values obtained from lung cancer cell lines were changed into numeric form after filtering the data. As IC50 values represent the degree of sensitivity toward drugs, they were considered as continuous regression targets. Nevertheless, as IC50 values can fluctuate greatly and can induce a skewed distribution, it was necessary to apply log10 transformation to the data prior to modeling.

Consequently, from this step, the value of the log10(IC50) was calculated for each cell line, which was used as the target variable for the Random Forest regression model. Log10(IC50) was computed with an intention of reducing the effects of high IC50 values on the model and balancing IC50 values on the scale of drug response values. Consequently, the use of log10(IC50) instead of IC50 was done throughout all analysis procedures.

2.5. Correlation-Based Feature Selection

Since the gene expression data set included many gene attributes, there was a need to reduce its dimensionality before performing modeling. In this regard, correlation-based attribute selection technique was implemented. The first step was to delete the depmap_id, raw IC50, and log10(IC50) attributes from the final dataset and create a matrix consisting of gene expression attributes only. Zero variance genes were further filtered out since these attributes do not vary among cell lines and cannot be used in predicting the response to drugs..

Correlation coefficient values were calculated for each gene using the Pearson correlation method between the expression value and the log10 of IC50. These correlations allowed for the determination of the level of linear association between each gene and the response to doxorubicin treatment. The value selected for the feature selection criterion was $|r| \geq 0.25$.

The above filtering process led to the selection of 457 genes. Genes with a positive correlation might have an increased log10(IC50) value indicating resistance to the drugs; genes with negative correlations might have decreased log10(IC50) values indicating increased sensitivity to the drugs. The 457 genes selected above were considered the input feature set for the baseline Random Forest model.

2.6. Baseline Random Forest Regression

Baseline Random Forest Regression was employed based on the identification of 457 genes using correlation-based feature selection. Random Forest is an ensemble learning technique that makes use of many decision trees [2]. It was considered a good fit for use in predicting drug response because of its capabilities in detecting non-linearities in high dimensional biological data [10].

In this model, the input variables included 457 gene expression features selected by correlation, and the target variable was the log10(IC50). In order to better assess the model performance, 5-fold cross validation was carried out. For that, the whole data set was randomly divided into five parts; in every round, four parts acted as the training sets and one part as the testing set.

The modeling process involved training of the model with 500 trees per fold. Prediction performance was measured using the three indicators: MAE, RMSE, and R^2. The first two were used as indicators of prediction error, while the latter was used to measure the explanatory ability of the predicted log10(IC50) and actual log10(IC50) data. In this study, R^2 was calculated as the squared Pearson correlation between actual and predicted log10(IC50) values. Finally, the result of modeling with the baseline gene set served as the benchmark for comparison with the RFE models.

2.7. Recursive Feature Elimination

Following the development of the Random Forest base model, Recursive Feature Elimination (RFE) technique was employed for deriving smaller and more relevant sets of genes [6]. RFE is a feature selection technique wherein features whose contribution towards building the model is lesser according to their feature importance values are iteratively eliminated from the model [6]. In this particular experiment, the %IncMSE importance measure from Random Forest was used to determine the relevance of the genes to the model.

RFE was carried out in two steps. First, with an initial selection of genes using correlation analysis involving 457 genes, a Random Forest method was applied in iterations to estimate gene importance levels. At each iteration, around 3% of those genes with the lowest importance level were eliminated. The process was repeated until the number of genes was decreased to 52. This step aimed at selecting a subset from the large set of genes.

In the second step, the left-out genes were analyzed further and just one gene having the least importance level was excluded in every iteration cycle. This resulted in the creation of fewer sets of genes, beginning with 52 genes. The gene set created in every iteration was documented and stored as a new candidate model.

It was because of this RFE method that it became feasible to evaluate the predictive ability of different gene sets. In other words, the goal was to choose an ultimate gene signature that would provide good prediction accuracy as well as high explainability.

2.8. 5-Fold Cross-Validation of RFE Gene Sets

Many candidate gene sets were identified through the RFE approach, which have different gene set sizes. Each of these candidate gene sets is tested as an individual random forest regression model. In order to compare the results fairly among all models, 5-fold cross validation is used.

In cross-validation, the data was split into five folds. Four folds were taken for training while the left-out fold served as the testing fold in every run. The entire process was carried out five times, during which all the folds were used once as the testing fold. In each case, Random Forest model training was performed using 500 trees for each gene set with the log10(IC50) prediction.

For each fold, calculations for MAE, RMSE, and R^2 were made. Afterward, the average value of Mean MAE, Mean RMSE, and Mean R^2 was computed for each gene set. Also, Overall MAE, Overall RMSE, and Overall R^2 values were computed based on log10(IC50) values from all folds using actual and predicted values. Such overall measures were used to compare overall prediction performance values of different RFE gene sets.

In this case, not only the selection of the model with the smallest number of genes, but also the determination of the best combination of genes which had the best performance in both prediction and explanation was the objective. This was done by comparing the results of RFE models in RMSE and R^2.

2.9. Final Model Selection

Following the results from RFE, the remaining candidate genes were validated using 5-fold cross-validation prior to carrying out the process of model selection. Two main parameters were used for comparing the models: Overall RMSE and Overall R^2. Overall RMSE is a measure of the error in predicting the data using the model. Meanwhile, Overall R^2 measures the explanatory power of the model in predicting the log10(IC50).

As per the findings, the 8-gene RFE model gave the lowest Overall RMSE score. The benefits of using this method were its accuracy and low prediction error since it uses fewer genes. Nevertheless, the 60-gene RFE model provided the highest Overall R^2 value. Higher values indicate a better ability to explain log10(IC50) variability.

For this reason, the predictive model consisting of 60 genes was selected as the final predictive signature. This decision took into account not only the prediction accuracy but also the biological relevance. The selection of the predictive signature was based on the consideration that the 60-gene model is more biologically relevant due to its relatively large gene list compared to the 8-gene model.

2.10. Functional Enrichment Analysis

The functional enrichment analysis was done to find the biological relevance of the Random Forest RFE model having 60 genes. For doing this, the g:Profiler web server was used, where the organism was specified as Homo sapiens [9]. The final gene list of 60 genes was taken as the input for the analysis.

For the enrichment analysis, Benjamini-Hochberg FDR adjustment method was used as the significance cutoff method, while the user threshold value was taken as 0.05. “Only annotated genes” was chosen as the statistical domain scope, while numeric ID evaluation was done for the ENTREZGENE_ACC identifier.

The analysis was carried out to identify those biological functions, cell compartments, or protein complex classes that the genes selected as part of the final 60-gene signature might be involved in. As a result, the analysis enabled the interpretation of the final predictive model not only through performance metrics but also biologically. In particular, the biological functions involved in localization, vesicles, and signaling were considered in evaluating the possible biological implications of the final list of genes.

3. RESULTS

3.1. Final Dataset

The final modeling dataset consisted of the CCLE expression data merged with the PRISM IC50 data of doxorubicin. In this process, first, cell lines having their IC50 values of doxorubicin were picked up from the PRISM dataset. Here, 471 such cell lines with doxorubicin IC50 values were found. After that, those cell lines were filtered using the DepMap ID values present in the CCLE gene expression data, which finally gave us 460 overlapping cell lines. The cancer-type filter narrowed down the number of cancer cell lines used in the study to only 96 for lung cancer. The model building process took place within the context of lung cancer. After the transformation of IC50 values, we have log10(IC50) as the final response variable. For all prediction analysis, we have considered the log10 (IC50) as our objective function. The gene expression matrix had 19,220 genes at first but after applying the correlation-based filtering, there were only 457 genes left. A summary of the dataset integration and preprocessing steps is shown in Table 3.1.1.

Table 3.1.1. Summary of Dataset Integration and Preprocessing

Processing Step	Criterion / Method	Output
Drug response filtering	PRISM entries for doxorubicin with available IC50 values	471 cell lines
Dataset matching	Common DepMap IDs between CCLE expression and PRISM IC50 data	460 cell lines
Cancer type filtering	Cell lines annotated as lung cancer in sample information	96 cell lines
Target transformation	IC50 values transformed as log10(IC50)	1 response variable
Initial gene expression matrix	Gene expression features before feature selection	19,220 genes
Correlation-based feature filtering	Genes with |Pearson correlation with log10(IC50)| $\geq$ 0.25	457 genes

3.2. Correlation-Based Feature Selection Results

Feature selection based on Pearson correlation identified gene expression features related to the log10 IC50 value. The values of pearson correlation coefficient for each gene demonstrated the nature and magnitude of the association between gene expression and drug sensitivity. It was observed that there existed a different nature and magnitude of associations between genes and drug sensitivity.

Figure 3.2.1 illustrates the correlation distribution between gene expression levels and log10(IC50) in lung cancer cell lines. The distribution mostly lies in areas near a correlation value of zero. This indicates that most of the genes are not significantly correlated with the response to doxorubicin treatment. A Pearson correlation cutoff of ≥ 0.25 was used in selecting the genes for log10(IC50).

Consequently, there were 457 genes that formed the basis of the Random Forest regression model used as the baseline. Positive correlation could mean that such genes tend to be more likely to have high log10(IC50), meaning that the cells exhibit more resistance to doxorubicin. In contrast, negative correlation could mean that genes have low log10(IC50), meaning more sensitivity towards doxorubicin. Thus, correlation-based feature selection gave an informative gene set to be considered in the next step.

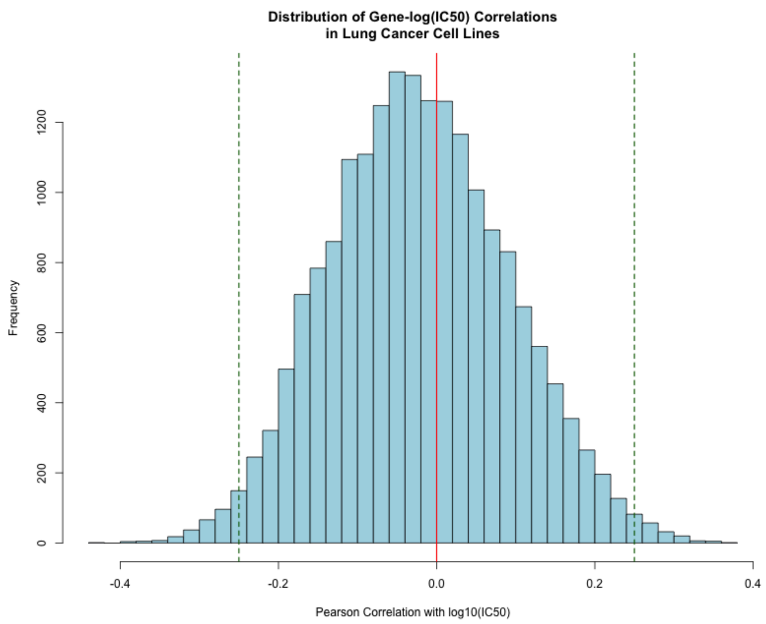


Figure 3.2.1. Distribution of gene–log10(IC50) correlations in lung cancer cell lines

3.3. Baseline Random Forest Performance

The baseline Random Forest regression model was built using the initial set of 457 genes selected based on correlations, showing baseline performance prior to RFE. The MAE, RMSE, and R^2 of the baseline model were obtained through 5-fold cross-validation.

The results of 5-fold cross-validation for the Baseline Random Forest model are shown in Table 3.3.1. Mean MAE, RMSE, and R^2 values obtained in 5 iterations were equal to 0.2264, 0.3021, and 0.2596 respectively. Thus, it is possible to conclude that the Baseline Random Forest model built on the basis of 457 selected genes shows some degree of prediction efficiency; however, its explanatory ability can be improved further.

Table 3.3.1. Baseline Random Forest 5-Fold Cross-Validation Results

Performance Metric	**Mean Across 5 Folds**	**Standard Deviation**
MAE	0.2264	0.0134
RMSE	0.3021	0.0223
R^2	0.2596	0.1317

The figure below, Figure 3.3.1, illustrates the actual vs. the predicted log10(IC50) obtained through the 5-fold cross-validation of the baseline model using Random Forests. It can be seen that the points lie partly along the line of perfect prediction. This means that the baseline model is predicting some of the log10(IC50) accurately but not all. The performance of this model can therefore be deemed acceptable but RFE tuning was performed to improve its performance.

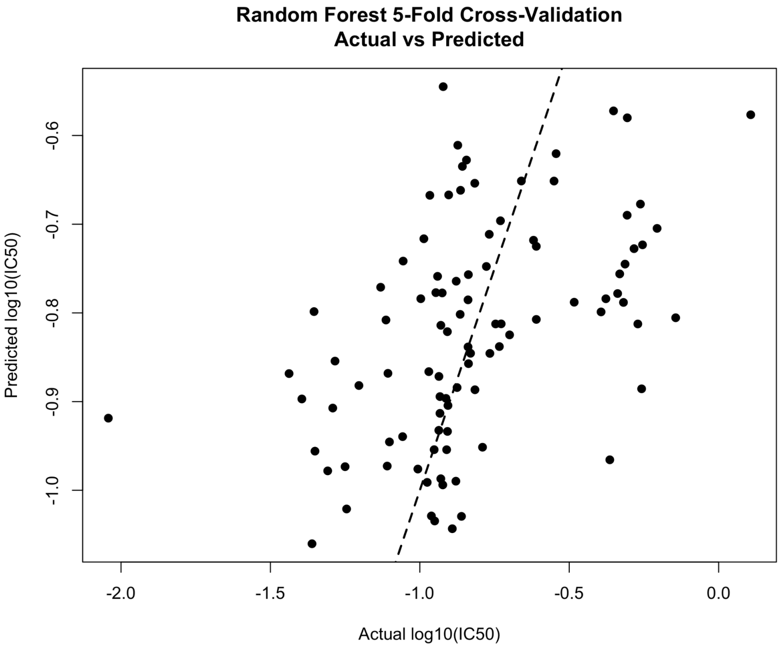


Figure 3.3.1. Baseline Random Forest actual versus predicted log10(IC50) values

3.4. RFE Model Optimization

The process of recursive feature elimination led to more effective and informative groups of genes when compared with the baseline model. The deletion of genes that had little impact using Random Forest feature importance measures enabled the selection of varying numbers of RFE gene sets. The results achieved from 5-fold cross-validation for each of the RFE gene sets were used for comparison.

Figure 3.4.1 depicts the Overall R2 values for various gene set sizes selected via the RFE method. From the table, it can be noted that some reduced gene sets using the RFE method had a higher value of R2 compared to the baseline model. The maximum value of Overall R2 was recorded for the 60-gene model. This indicates that the 60-gene model best explains the relationship.

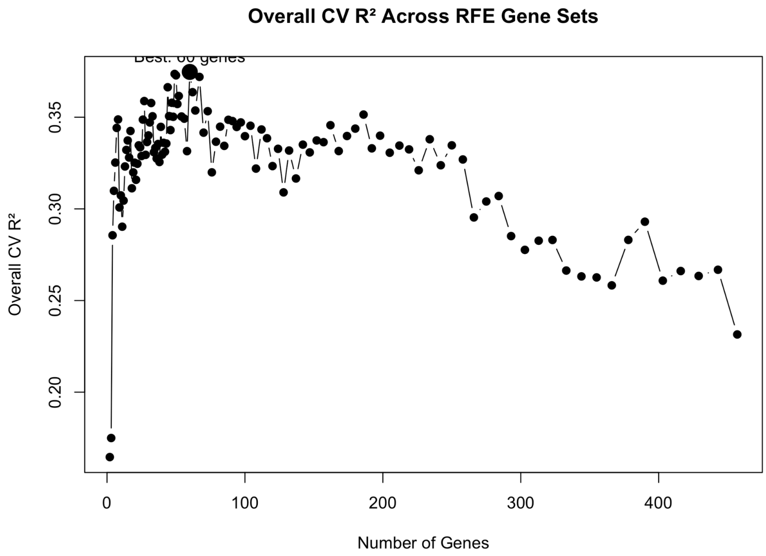


Figure 3.4.1. Overall R^2 across RFE-selected Random Forest gene sets

As illustrated in Figure 3.4.2, there is a variance of the overall RMSE depending on the size of the gene set selected via RFE. In terms of the RMSE, we observed that the smaller gene sets were able to improve the prediction error compared with the baseline model. The minimum Overall RMSE was recorded in the 8-gene set. Nonetheless, the overall RMSE for the 60-gene set was still very close to the 8-gene set.

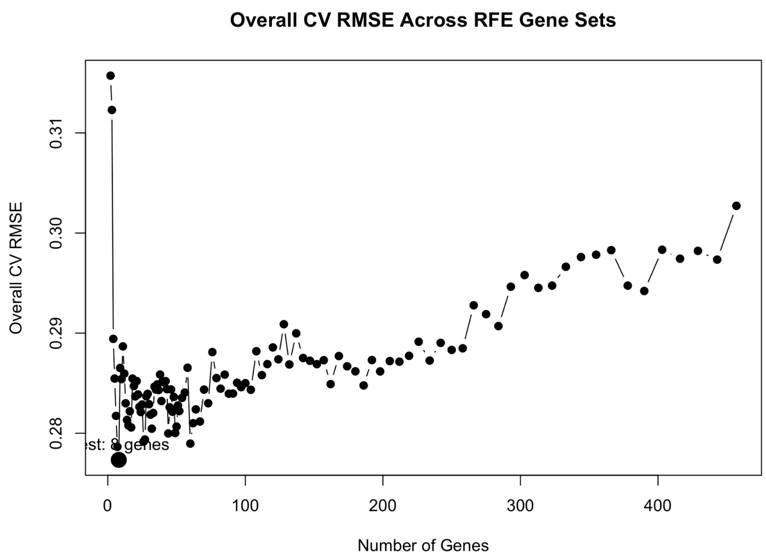


Figure 3.4.2. Overall RMSE across RFE-selected Random Forest gene sets

The performance evaluation of the baseline model, optimal RMSE RFE model, and final chosen RFE model is shown in Table 3.4.1. The Overall RMSE for the baseline Random Forest model containing 457 genes is 0.3027 with an Overall R^2 of 0.2315. For the 8-gene model using RFE, the Overall RMSE is lowered to 0.2773, with the minimum prediction error. However, for the 60-gene model, the highest explanatory performance is observed, with an Overall R^2 of 0.3747.

For this reason, the 60-gene model via RFE algorithm was selected to be used in the final model selection step. Despite being smaller with a reduced RMSE, the 60-gene model performed better in terms of Overall R2 value and could provide a larger biological field of interpretation.

Table 3.4.1. Performance Comparison of Baseline and RFE-Optimized Random Forest Models

Model	**Gene Set Size**	**Mean RMSE**	**Mean R^2**	**Overall RMSE**	**Overall R^2**	**Selection Reason**
Baseline Random Forest	457 genes	0.3021	0.2596	0.3027	0.2315	Initial model after correlation-based filtering
Best RMSE RFE model	8 genes	0.2760	0.3926	0.2773	0.3488	Lowest overall RMSE
Final selected RFE model	60 genes	0.2786	0.4071	0.2790	0.3747	Highest overall R^2

3.5. Final 60-Gene Predictive Signature

As per the outcome of the optimization of RFE model, the 60-gene Random Forest RFE model is considered as the chosen predictive signature. The first reason for choosing this model as the final model was due to its having the highest value of Overall R^2. As can be seen from Table 3.4.1, the value of Overall R^2 obtained by the 60-gene Random Forest RFE model is 0.3747, which is more explanatory than the baseline Random Forest model.

Despite the low Overall RMSE score obtained using the 8-gene RFE algorithm, its gene set was smaller than that obtained from the 60-gene model in terms of functional enrichment and biological relevance. On the other hand, the 60-gene model yielded better predictive power while giving a more inclusive gene set, which is suitable for biological purposes.

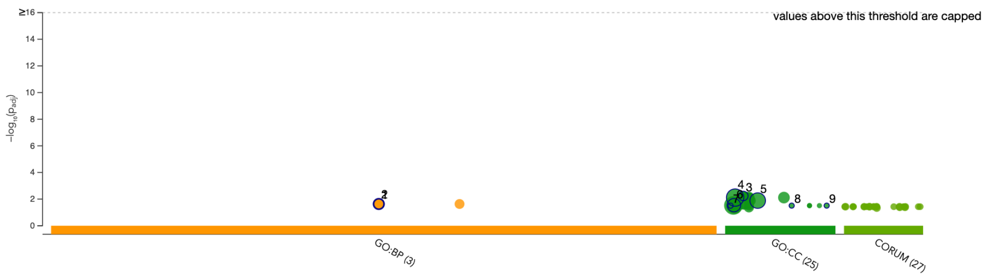
Genes involved in the predictive signature of 60 genes include GRN, NUP98, MAP2K2, KLF4, RBM39, EEF2, and WIPI2, and they are known to be related either with chemotherapeutic response or drug resistance. Genes identified in the predictive signature have biological relevance in relation to the effects of drugs, including gene regulation involved in apoptosis, nuclear transport, MAP kinase pathway, transcriptional control, RNA splicing, protein synthesis, and autophagy. Among the final genes, NUP98 has previously been reported as a predictor of response to anthracycline-based chemotherapy, supporting its possible relevance to anthracycline response biology [8]. These findings show that the final signature is biologically meaningful in addition to being statistically relevant.

The complete list of genes included in the final 60-gene predictive signature is provided in Appendix Table A1.

3.6. Enrichment Analysis Results

In order to perform functional enrichment analysis on the 60-gene Random Forest RFE signature, g:Profiler was used. The species that was chosen was Homo sapiens. For the significance threshold, Benjamini-Hochberg FDR correction was used, with a user threshold of 0.05. The scope was set to “only annotated genes” and the numeric ID evaluation was done based on ENTREZGENE_ACC.

The outcomes of the g:Profiler enrichment analysis are illustrated in Figure 3.6.1. Outcomes of enrichment were majorly found in the GO cell components groupings. Some of the terms that included cytosol, organelle sub-compartments, late endosomes, trans-Golgi network transport vesicle membranes, and granular vesicles were identified. Furthermore, another term, Grb2-Sos complex, related to signaling associated protein complexes was also found in the CORUM grouping.



ID	Source	Term ID	Term Name	p_{adj} (query_1)
1	GO:BP	GO:0048679	regulation of axon regeneration	2.325×10^{-2}
2	GO:BP	GO:0048678	response to axon injury	2.325×10^{-2}
3	GO:CC	GO:0012510	trans-Golgi network transport vesicle membrane	5.771×10^{-3}
4	GO:CC	GO:0005829	cytosol	7.680×10^{-3}
5	GO:CC	GO:0031984	organelle subcompartment	1.293×10^{-2}
6	GO:CC	GO:0005770	late endosome	2.855×10^{-2}
7	GO:CC	GO:0005582	collagen type XV trimer	3.086×10^{-2}
8	GO:CC	GO:0070618	Grb2-Sos complex	3.086×10^{-2}
9	GO:CC	GO:1990005	granular vesicle	3.086×10^{-2}

version	e114_eg62_p19_27110d83
date	09.06.2026 17:39:16
organism	hsapiens

g:Profiler

Figure 3.6.1. g:Profiler enrichment analysis of the final 60-gene signature

The findings point to the conclusion that the final 60-gene signature does not represent a single drug-response pathway. On the contrary, the gene signature can reflect a transcriptional signature related to cellular organization, organelles, endo-lysosomal traffic, and other signal-related cellular components. These observations suggest that doxorubicin response may occur through multiple biological processes involving resistance-related mechanisms.

The appearance of autophagy-related genes such as WIPI2, MEK/ERK signaling-related genes such as MAP2K2, protein translation-related genes like EEF2, and survival/apoptosis genes such as GRN in the list of final genes, along with terms related to intracellular organization and vesicles in the enrichment analysis, indicates that the final signature has relevance to drug response biology.

4. DISCUSSION

For this work, the gene expression data from CCLE and IC50 data from PRISM were combined to perform a prediction of doxorubicin response on lung cancer cell lines through machine learning algorithms. The final dataset contained 96 cell lines of lung cancer. Considering the high dimensionality of the gene expression data, we performed gene selection using feature selection methods based on the Pearson correlation coefficient for 457 genes that were related to the log10 (IC50). These genes were used in the baseline Random Forest Regression algorithm.

Performance of the Baseline Random Forest model resulted in mean RMSE = 0.3021 and mean R^2 = 0.2596. Performance results demonstrated that the Baseline model has Overall RMSE = 0.3027 and Overall R^2 = 0.2315. From this information, it can be concluded that among the genes selected based on correlations there is a specific degree of predictability in terms of doxorubicin resistance. The small value of R^2 means that not all selected 457 genes are needed for the model, and optimization is required. Therefore, RFE using random forest importance value was employed. RFE models were evaluated for various sets of genes using 5-fold cross-validation. Consequently, the improvement was found in prediction error and explanation in comparison with the baseline model thanks to RFE optimization. RFE with 8 genes was determined as having the smallest Overall RMSE value and the most concise among other RFE models. However, the 60-gene RFE model was selected as final owing to its highest Overall R^2 value. Overall increase in R^2 from 0.2315 to 0.3747 means that RFE was able to find an informative subset of genes.

Choosing the final 60-gene model took into account not only statistical measures, but also the potential biological interpretation of the results. Although there were obvious benefits to the 8-gene model in terms of error, the extremely small number of genes may affect the ability of this gene set to be functionally enriched and interpreted biologically. On the other hand, the 60-gene model had a lower RMSE than the baseline model as well as the largest Overall R^2. Some of the 60 genes selected in the final 60-gene signature were considered to be related to chemotherapy resistance/response mechanisms. For instance, the gene GRN is an example of a gene that is involved in survival signaling and apoptosis [12]. Gene NUP98 is an example of a gene that can be discussed in relation to nuclear transport and anthracycline resistance biomarkers [8]. The MAP2K2 gene is responsible for the MEK/ERK signaling pathway, which might be related to cell survival and regulation of drug transporters and resistance signaling pathways [3]. Another one is KLF4, which is a crucial transcription factor that plays a role in transcriptional regulation, stemness properties, EMT processes, and chemotherapy resistance mechanisms. Finally, another one would be RBM39, which can be examined in relation to RNA splicing and targeted drug resistance/response [7].

Results from functional enrichment analysis revealed that the resulting 60-gene signature was not biased toward any one primary doxorubicin resistance pathway. Rather, the terms enriched had higher representation among cellular components, including cytosol, organelle subcompartment, late endosome, trans-Golgi network, transport vesicle membrane, and granular vesicle. These findings suggest that the response of cells to doxorubicin is not biased toward one single pathway, such as apoptosis or DNA damage response, but rather is involved in more general cellular processes.

This finding shows that there is the need for assessment of transcriptomic signatures in studies on predicting drug responses based on different gene patterns as opposed to individual genes. Resistance to doxorubicin is a multigenic process that may involve several biological processes, such as drug efflux, inhibition of apoptosis, DNA repair, autophagy, oxidative stress response, and survival signaling. This explains why various genes associated with different biological pathways were used in the selection of the 60-gene signature.

The present work also has several disadvantages. First, the number of samples in the dataset was reduced to 96 due to lung cancer filtering. Such a low number of samples is rather insufficient for performing modeling of gene expression. Second, the development of the model was performed on the basis of cell line data. Thus, it can be considered incorrect to state that the proposed method will be directly applicable to tumor samples. Third, validation on an external dataset was not performed in order to test the final signature. The resulting set of 60 genes must be considered merely a candidate signature.

Overall, it can be concluded from this research that large-scale pharmacogenomic databases like CCLE and PRISM could be combined and employed for the development of personalized cancer drug response prediction models. The developed 60-gene RF model selected via the RFE approach was found more efficient than the baseline model and had biologically relevant genes. Further research into this topic may include external validation, alternative algorithms, and experiments.

5. CONCLUSION

For this analysis, CCLE RNAseq and PRISM IC50 datasets have been combined to identify the predictors of doxorubicin resistance in lung cancer cell lines. The IC50 values have been transformed into log10 IC50 values and used as continuous target variables for Random Forest regression. Using Pearson correlation-based feature selection, 457 genes have been chosen.

Comparing different sizes of gene sets in RFE optimization led to the choice of the 60-gene model as the best predictive model due to its maximum Overall R^2 value. The model exhibited a lower RMSE value and a higher R^2 value compared to the baseline Random Forest model. Functional enrichment analysis further indicated that the final signature is enriched in genes related to intracellular organization, vesicles, lysosomes/endosomes, and signaling-associated parts of cells.

The resulting 60-gene signature consists of genes possibly related to drug resistance or chemoresistance-related mechanisms, like GRN, NUP98, MAP2K2, KLF4, RBM39, EEF2, and WIPI2. This indicates that the doxorubicin response may represent a multifactorial phenomenon governed by a wide range of biological pathways. Overall, this study shows that the 60-gene signature based on the random forest feature selection technique can be suggested as a transcriptomic biomarker for doxorubicin response in lung cancer cells. Nevertheless, further independent validation experiments are required to confirm its applicability.

REFERENCES

[1] Barretina, J., Caponigro, G., Stransky, N., Venkatesan, K., Margolin, A. A., Kim, S., Wilson, C. J., Lehár, J., Kryukov, G. V., Sonkin, D., Reddy, A., Liu, M., Murray, L., Berger, M. F., Monahan, J. E., Morais, P., Meltzer, J., Korejwa, A., Jané-Valbuena, J., Garraway, L. A. (2012). Addendum: The Cancer Cell Line Encyclopedia enables predictive modelling of anticancer drug sensitivity. *Nature*, *492*(7428), 290. https://doi.org/10.1038/nature11735

[2] Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/a:1010933404324

[3] Caunt, C. J., Sale, M. J., Smith, P. D., & Cook, S. J. (2015). MEK1 and MEK2 inhibitors and cancer therapy: the long and winding road. *Nature Reviews. Cancer*, *15*(10), 577–592. https://doi.org/10.1038/nrc4000

[4] Corsello, S. M., Nagari, R. T., Spangler, R. D., Rossen, J., Kocak, M., Bryan, J. G., Humeidi, R., Peck, D., Wu, X., Tang, A. A., Wang, V. M., Bender, S. A., Lemire, E., Narayan, R., Montgomery, P., Ben-David, U., Garvie, C. W., Chen, Y., Rees, M. G., Golub, T. R. (2020). Discovering the anticancer potential of non-oncology drugs by systematic viability profiling. *Nature Cancer*, *1*(2), 235–248. https://doi.org/10.1038/s43018-019-0018-6

[5] Dong, Z., Zhang, N., Li, C., Wang, H., Fang, Y., Wang, J., & Zheng, X. (2015). Anticancer drug sensitivity prediction in cell lines from baseline gene expression through recursive feature selection. *BMC Cancer*, *15*(1), 489. https://doi.org/10.1186/s12885-015-1492-6

[6] Guyon, I., Weston, J., Barnhill, S., & Vapnik, V. (2002). Gene Selection for Cancer Classification using Support Vector Machines. *Machine Learning*, *46*(1–3), 389–422. https://doi.org/10.1023/a:1012487302797

[7] Han, T., Goralski, M., Gaskill, N., Capota, E., Kim, J., Ting, T. C., Xie, Y., Williams, N. S., & Nijhawan, D. (2017). Anticancer sulfonamides target splicing by inducing RBM39 degradation via recruitment to DCAF15. *Science*, *356*(6336). https://doi.org/10.1126/science.aal3755

[8] Mullan, P. B., Bingham, V., Haddock, P., Irwin, G. W., Kay, E., McQuaid, S., & Buckley, N. E. (2019). NUP98 – a novel predictor of response to anthracycline-based chemotherapy in triple negative breast cancer. *BMC Cancer*, *19*(1), 236. https://doi.org/10.1186/s12885-019-5407-9

[9] Raudvere, U., Kolberg, L., Kuzmin, I., Arak, T., Adler, P., Peterson, H., & Vilo, J. (2019). *g:Profiler: a web server for functional enrichment analysis and conversions of gene lists*. Nucleic Acids Research, 47(W1), W191–W198. https://doi.org/10.1093/nar/gkz369

[10] Riddick, G., Song, H., Ahn, S., Walling, J., Borges-Rivera, D., Zhang, W., & Fine, H. A. (2010). Predicting in vitro drug sensitivity using Random Forests. *Bioinformatics*, *27*(2), 220–224. https://doi.org/10.1093/bioinformatics/btq628

[11] Siegel, R. L., Giaquinto, A. N., & Jemal, A. (2024). Cancer statistics, 2024. *CA a Cancer Journal for Clinicians*, *74*(1), 12–49. https://doi.org/10.3322/caac.21820

[12] Zhou, C., Huang, Y., Wu, J., Wei, Y., Chen, X., Lin, Z., & Nie, S. (2021). A narrative review of multiple mechanisms of progranulin in cancer: a potential target for anti-cancer therapy. *Translational Cancer Research*, *10*(9), 4207–4216. https://doi.org/10.21037/tcr-20-2972

APPENDIX

Appendix Table A1. Final 60-Gene Predictive Signature Selected by the Random Forest RFE Model

No.	Gene Symbol	No.	Gene Symbol	No.	Gene Symbol
1	GRN	21	RBM39	41	ARMC4
2	CLTCL1	22	CSNK1G2	42	ZNF212
3	CRYBB3	23	SWAP70	43	SLC23A1
4	SGK2	24	AP4B1	44	SNAPC5
5	MYL9	25	ELP2	45	PTDSS2
6	MINDY4	26	KLF4	46	DSG4
7	ZNHIT1	27	TINAG	47	EIF3F
8	CNTNAP1	28	GPAT3	48	MTURN
9	CARS1	29	ARSG	49	AP2A2
10	NUP98	30	AZIN2	50	NANOS3
11	EHBP1	31	CAND2	51	LCTL
12	PHTF1	32	SHISAL2B	52	ASTL
13	ZCCHC17	33	SSBP3	53	OR2T3
14	KHDRBS1	34	WIPI2	54	AKR1C4
15	CCZ1	35	HPD	55	CHML
16	MXD4	36	EMC10	56	COL15A1
17	CPNE5	37	RPL27A	57	LRRC69
18	FLRT3	38	CATSPER2	58	SHISA9
19	MAP2K2	39	SUN5	59	AP5Z1
20	CHTF18	40	EEF2	60	POM121C

```
############################################################
# DATASET INTEGRATION: CCLE EXPRESSION AND PRISM IC50
############################################################

# 1. set the working directory
setwd("~/Downloads/DepMap_Project")   # change accordingly
getwd()

# 2. load the required package
library(data.table)

# 3. load datasets
expr <- fread("CCLE_expression.csv")
prism <-
fread("prism-repurposing-20q2-secondary-screen-dose-response-curve-parameters.csv")
sample_info <- fread("sample_info.csv")

# 4. filter doxorubicin
doxo <- prism[name == "doxorubicin"]    # drug names are lowercase

# remove missing IC50 values
doxo <- doxo[!is.na(ic50)]

# check number of cell lines
nrow(doxo)

# 5. prepare expression dataset (rename first column to depmap_id)
setnames(expr, "V1", "depmap_id")

# 6. identify common cell lines
common_ids <- intersect(expr$depmap_id, doxo$depmap_id)
length(common_ids)

# 7. subset the expression dataset to keep only cell lines that have matching IC50 data
expr_subset <- expr[
  depmap_id %in% common_ids
]

# 8. subset IC50 data
doxo_subset <- doxo[
  depmap_id %in% common_ids,
  .(depmap_id, ic50)
]
```

```
# 9. merge expression and IC50
final_dataset <- merge(
  expr_subset,
  doxo_subset,
  by = "depmap_id"
)

# 10. filter lung cancer cell lines using primary_disease
lung_ids <- sample_info[
  grepl("lung", primary_disease, ignore.case = TRUE),
  DepMap_ID
]

# 11. keep only lung cancer samples in the final dataset
final_dataset <- final_dataset[
  depmap_id %in% lung_ids
]

# 12. check dataset dimensions after filtering
dim(final_dataset)
head(final_dataset)

# 13. convert IC50 to numeric (IC50 may be stored as character → convert to numeric)
final_dataset$ic50 <- as.numeric(final_dataset$ic50)

# 14. Log-transform IC50 (Log transformation improves regression modeling)
final_dataset$log_ic50 <- log10(final_dataset$ic50)

# 15. save Dataset A
fwrite(final_dataset, "Dataset_A_gene_expression_ic50.csv")

################################################################

################################################################
# CORRELATION-BASED FEATURE SELECTION
################################################################

# 1. set the working directory
setwd("~/Downloads/DepMap_Project")

# 2. load the required package
library(data.table)
```

```
# 3. load datasets
final_dataset <- fread("Dataset_A_gene_expression_ic50.csv")

# 4. remove zero-variance genes
X <- final_dataset[, !c("depmap_id","ic50","log_ic50"), with=FALSE]
X <- X[, sapply(X, sd, na.rm=TRUE) > 0, with=FALSE]

# 5. correlation
cor_results <- cor(X, final_dataset$log_ic50, method="pearson")

# 6. convert correlation results to a vector
cor_vec <- cor_results[, 1]

# 7. plot distribution of correlation values
png("lung_gene_correlation_distribution.png", width = 800, height = 650)

hist(cor_vec,
    breaks = 50,
    main = "Distribution of Gene-log(IC50) Correlations\nin Lung Cancer Cell Lines",
    xlab = "Pearson Correlation with log10(IC50)",
    col = "lightblue",
    border = "black")

# add threshold lines (+0.25 and -0.25) and zero reference line
abline(v = 0.25, col = "darkgreen", lwd = 1.5, lty = 2)
abline(v = -0.25, col = "darkgreen", lwd = 1.5, lty = 2)
abline(v = 0, col = "red", lwd = 1.5)

dev.off()

# 8. select genes with |correlation| >= 0.25 (threshold-based filtering)
selected_genes <- cor_vec[abs(cor_vec) >= 0.25]

# 9. number of selected genes
length(selected_genes)

# 10. create initial feature matrix using selected genes
X_filtered <- X[, names(selected_genes), with = FALSE]

# 11. check dimensions
dim(X_filtered)
```

```
# 12. save filtered matrix
fwrite(X_filtered, "filtered_gene_matrix.csv")

# 13. save selected gene names
write.csv(names(selected_genes), "selected_genes.csv", row.names = FALSE)

################################################################

################################################################
# BASELINE RANDOM FOREST 5-FOLD CROSS-VALIDATION
# USING 457 CORRELATION-SELECTED GENES
################################################################

# 1. set the working directory
setwd("~/Downloads/DepMap_Project")

# 2. load the required packages
library(data.table)
library(randomForest)

# 3. load processed datasets
final_dataset <- fread("Dataset_A_gene_expression_ic50.csv")
X <- fread("filtered_gene_matrix.csv")

# 4. define target variable
y <- final_dataset$log_ic50

# 5. convert feature matrix to data frame
X <- as.data.frame(X)

# 6. check dimensions
cat("Number of samples =", nrow(X), "\n")
cat("Number of selected genes =", ncol(X), "\n")

# 7. create metric function
calculate_metrics <- function(actual, predicted) {

  mae <- mean(abs(predicted - actual))
  rmse <- sqrt(mean((predicted - actual)^2))
  r2 <- cor(predicted, actual)^2

  return(c(MAE = mae, RMSE = rmse, R2 = r2))
}
```

```
# 8. create 5 folds
set.seed(42)

k <- 5

folds <- sample(
  rep(1:k, length.out = nrow(X))
)

# 9. create empty data frames
cv_results <- data.frame(
  Fold = integer(),
  MAE = numeric(),
  RMSE = numeric(),
  R2 = numeric()
)

cv_predictions <- data.frame(
  Actual = numeric(),
  Predicted = numeric(),
  Fold = integer()
)

# 10. run 5-fold cross-validation
for (i in 1:k) {

  cat("Running fold", i, "\n")

  test_idx <- which(folds == i)
  train_idx <- setdiff(1:nrow(X), test_idx)

  X_train_cv <- X[train_idx, ]
  X_test_cv  <- X[test_idx, ]

  y_train_cv <- y[train_idx]
  y_test_cv  <- y[test_idx]

  set.seed(42)

  rf_cv_model <- randomForest(
    x = X_train_cv,
    y = y_train_cv,
    ntree = 500
```

```
  )

  pred_cv <- predict(
    rf_cv_model,
    X_test_cv
  )

  metrics_cv <- calculate_metrics(
    actual = y_test_cv,
    predicted = pred_cv
  )

  cv_results <- rbind(
    cv_results,
    data.frame(
      Fold = i,
      MAE = as.numeric(metrics_cv["MAE"]),
      RMSE = as.numeric(metrics_cv["RMSE"]),
      R2 = as.numeric(metrics_cv["R2"])
    )
  )

  cv_predictions <- rbind(
    cv_predictions,
    data.frame(
      Actual = y_test_cv,
      Predicted = pred_cv,
      Fold = i
    )
  )
}

# 11. reset row names
rownames(cv_results) <- NULL
rownames(cv_predictions) <- NULL

# 12. print fold results
print(cv_results)

# 13. calculate summary
cv_summary <- data.frame(
  Metric = c("MAE", "RMSE", "R2"),
  Mean = c(
```

```
    mean(cv_results$MAE),
    mean(cv_results$RMSE),
    mean(cv_results$R2)
  ),
  SD = c(
    sd(cv_results$MAE),
    sd(cv_results$RMSE),
    sd(cv_results$R2)
  )
)

print(cv_summary)

# 14. calculate overall cross-validated performance
overall_cv_mae <- mean(
  abs(cv_predictions$Predicted - cv_predictions$Actual)
)

overall_cv_rmse <- sqrt(
  mean((cv_predictions$Predicted - cv_predictions$Actual)^2)
)

overall_cv_r2 <- cor(
  cv_predictions$Predicted,
  cv_predictions$Actual
)^2

cat("Overall CV MAE =", overall_cv_mae, "\n")
cat("Overall CV RMSE =", overall_cv_rmse, "\n")
cat("Overall CV R² =", overall_cv_r2, "\n")

# 15. save cross-validation results
write.csv(
  cv_results,
  "rf_5fold_cv_results.csv",
  row.names = FALSE
)

write.csv(
  cv_summary,
  "rf_5fold_cv_summary.csv",
  row.names = FALSE
)
```

```
write.csv(
  cv_predictions,
  "rf_5fold_cv_predictions.csv",
  row.names = FALSE
)

# 16. save actual vs predicted plot
png(
  "rf_5fold_cv_actual_vs_predicted.png",
  width = 2400,
  height = 2000,
  res = 300
)

plot(
  cv_predictions$Actual,
  cv_predictions$Predicted,
  pch = 19,
  xlab = "Actual log10(IC50)",
  ylab = "Predicted log10(IC50)",
  main = "Random Forest 5-Fold Cross-Validation\nActual vs Predicted"
)

abline(
  0,
  1,
  lwd = 2,
  lty = 2
)

dev.off()

# 17. save RMSE by fold plot
png(
  "rf_5fold_cv_RMSE_by_fold.png",
  width = 2200,
  height = 1600,
  res = 300
)

barplot(
  cv_results$RMSE,
  names.arg = cv_results$Fold,
```

```
  xlab = "Fold",
  ylab = "RMSE",
  main = "Random Forest 5-Fold Cross-Validation RMSE by Fold"
)

dev.off()

#################################################################

#################################################################
# RANDOM FOREST RFE WITH 5-FOLD CV FOR ALL GENE SETS
#################################################################

setwd("~/Downloads/DepMap_Project")

library(data.table)
library(randomForest)

final_dataset <- fread("Dataset_A_gene_expression_ic50.csv")
X <- fread("filtered_gene_matrix.csv")

y <- final_dataset$log_ic50
X <- as.data.table(X)

cat("Number of samples =", nrow(X), "\n")
cat("Number of selected genes =", ncol(X), "\n")

calculate_metrics <- function(actual, predicted) {
  mae <- mean(abs(predicted - actual))
  rmse <- sqrt(mean((predicted - actual)^2))
  r2 <- cor(predicted, actual)^2

  return(c(MAE = mae, RMSE = rmse, R2 = r2))
}

#################################################################
# PART 1: CREATE RFE GENE SETS
#################################################################

set.seed(42)

train_idx <- sample(
  1:nrow(X),
```

```
  size = 0.8 * nrow(X)
)

X_train <- X[train_idx, ]
X_test  <- X[-train_idx, ]

y_train <- y[train_idx]
y_test  <- y[-train_idx]

current_genes <- colnames(X_train)

rfe_results <- data.frame(
  Stage = character(),
  iteration = integer(),
  n_genes = integer(),
  MAE = numeric(),
  RMSE = numeric(),
  R2 = numeric()
)

gene_list <- list()

################################################################
# PART 2: RFE BY REMOVING LOWEST 3% IMPORTANT GENES
################################################################

iteration <- 1

while (length(current_genes) > 52) {

  cat(
    "3% RFE iteration:",
    iteration,
    "| Number of genes:",
    length(current_genes),
    "\n"
  )

  set.seed(42)

  rf_model <- randomForest(
    x = X_train[, current_genes, with = FALSE],
    y = y_train,
```

```
  ntree = 500,
  importance = TRUE
)

pred <- predict(
  rf_model,
  X_test[, current_genes, with = FALSE]
)

metrics <- calculate_metrics(
  actual = y_test,
  predicted = pred
)

rfe_results <- rbind(
  rfe_results,
  data.frame(
    Stage = "RFE_3_percent",
    iteration = iteration,
    n_genes = length(current_genes),
    MAE = as.numeric(metrics["MAE"]),
    RMSE = as.numeric(metrics["RMSE"]),
    R2 = as.numeric(metrics["R2"])
  )
)

gene_list[[paste0("RFE_3_percent_", iteration)]] <- current_genes

importance_vals <- importance(rf_model)

importance_df <- data.frame(
  Gene = rownames(importance_vals),
  IncMSE = importance_vals[, "%IncMSE"]
)

importance_df <- importance_df[
  order(importance_df$IncMSE),
]

remove_n <- ceiling(
  0.03 * nrow(importance_df)
)
```

```
  genes_to_remove <- importance_df$Gene[1:remove_n]

  current_genes <- setdiff(
    current_genes,
    genes_to_remove
  )

  iteration <- iteration + 1
}

##############################################################
# PART 3: REMOVE REMAINING GENES ONE BY ONE
##############################################################

single_iteration <- 1

while (length(current_genes) > 1) {

  cat(
    "Single-gene elimination iteration:",
    single_iteration,
    "| Number of genes:",
    length(current_genes),
    "\n"
  )

  set.seed(42)

  rf_model <- randomForest(
    x = X_train[, current_genes, with = FALSE],
    y = y_train,
    ntree = 500,
    importance = TRUE
  )

  pred <- predict(
    rf_model,
    X_test[, current_genes, with = FALSE]
  )

  metrics <- calculate_metrics(
    actual = y_test,
    predicted = pred
```

```
  )

  rfe_results <- rbind(
    rfe_results,
    data.frame(
      Stage = "Single_gene_elimination",
      iteration = single_iteration,
      n_genes = length(current_genes),
      MAE = as.numeric(metrics["MAE"]),
      RMSE = as.numeric(metrics["RMSE"]),
      R2 = as.numeric(metrics["R2"])
    )
  )

  gene_list[[paste0("Single_gene_elimination_", single_iteration)]] <- current_genes

  importance_vals <- importance(rf_model)

  importance_df <- data.frame(
    Gene = rownames(importance_vals),
    IncMSE = importance_vals[, "%IncMSE"]
  )

  importance_df <- importance_df[
    order(importance_df$IncMSE),
  ]

  gene_to_remove <- importance_df$Gene[1]

  current_genes <- setdiff(
    current_genes,
    gene_to_remove
  )

  single_iteration <- single_iteration + 1
}

#################################################################
# PART 4: 5-FOLD CV FUNCTION FOR ONE GENE SET
#################################################################

run_rf_cv_for_gene_set <- function(X_data, y_data, gene_set, model_name) {
```

```
set.seed(42)

k <- 5

folds <- sample(
  rep(1:k, length.out = nrow(X_data))
)

cv_results <- data.frame(
  Model = character(),
  Fold = integer(),
  n_genes = integer(),
  MAE = numeric(),
  RMSE = numeric(),
  R2 = numeric()
)

cv_predictions <- data.frame(
  Model = character(),
  Actual = numeric(),
  Predicted = numeric(),
  Fold = integer()
)

for (i in 1:k) {

  cat("Running", model_name, "- Fold", i, "\n")

  test_idx <- which(folds == i)
  train_idx <- setdiff(1:nrow(X_data), test_idx)

  X_train_cv <- X_data[train_idx, gene_set, with = FALSE]
  X_test_cv  <- X_data[test_idx, gene_set, with = FALSE]

  y_train_cv <- y_data[train_idx]
  y_test_cv  <- y_data[test_idx]

  set.seed(42)

  rf_cv_model <- randomForest(
    x = X_train_cv,
    y = y_train_cv,
    ntree = 500
```

```
  )

  pred_cv <- predict(
    rf_cv_model,
    X_test_cv
  )

  metrics_cv <- calculate_metrics(
    actual = y_test_cv,
    predicted = pred_cv
  )

  cv_results <- rbind(
    cv_results,
    data.frame(
      Model = model_name,
      Fold = i,
      n_genes = length(gene_set),
      MAE = as.numeric(metrics_cv["MAE"]),
      RMSE = as.numeric(metrics_cv["RMSE"]),
      R2 = as.numeric(metrics_cv["R2"])
    )
  )

  cv_predictions <- rbind(
    cv_predictions,
    data.frame(
      Model = model_name,
      Actual = y_test_cv,
      Predicted = pred_cv,
      Fold = i
    )
  )
}

overall_mae <- mean(
  abs(cv_predictions$Predicted - cv_predictions$Actual)
)

overall_rmse <- sqrt(
  mean((cv_predictions$Predicted - cv_predictions$Actual)^2)
)
```

```
  overall_r2 <- cor(
    cv_predictions$Predicted,
    cv_predictions$Actual
  )^2

  cv_summary <- data.frame(
    Model = model_name,
    n_genes = length(gene_set),
    Mean_MAE = mean(cv_results$MAE),
    Mean_RMSE = mean(cv_results$RMSE),
    Mean_R2 = mean(cv_results$R2),
    Overall_MAE = overall_mae,
    Overall_RMSE = overall_rmse,
    Overall_R2 = overall_r2
  )

  return(
    list(
      fold_results = cv_results,
      predictions = cv_predictions,
      summary = cv_summary
    )
  )
}

############################################################
# PART 5: RUN 5-FOLD CV FOR ALL RFE GENE SETS
############################################################

all_cv_summary <- data.frame()
all_cv_folds <- data.frame()
all_cv_predictions <- data.frame()

for (gene_set_name in names(gene_list)) {

  gene_set <- gene_list[[gene_set_name]]

  cat(
    "Running 5-fold CV for",
    gene_set_name,
    "| Number of genes:",
    length(gene_set),
    "\n"
```

```
  )

  cv_output <- run_rf_cv_for_gene_set(
    X_data = X,
    y_data = y,
    gene_set = gene_set,
    model_name = gene_set_name
  )

  all_cv_summary <- rbind(
    all_cv_summary,
    cv_output$summary
  )

  all_cv_folds <- rbind(
    all_cv_folds,
    cv_output$fold_results
  )

  all_cv_predictions <- rbind(
    all_cv_predictions,
    cv_output$predictions
  )
}

############################################################
# PART 6: SELECT BEST MODELS
############################################################

best_rmse_row <- which.min(all_cv_summary$Overall_RMSE)
best_r2_row <- which.max(all_cv_summary$Overall_R2)

best_rmse_model <- all_cv_summary[best_rmse_row, ]
best_r2_model <- all_cv_summary[best_r2_row, ]

cat("\nBest model by Overall RMSE:\n")
print(best_rmse_model)

cat("\nBest model by Overall R2:\n")
print(best_r2_model)

############################################################
# PART 7: SAVE RESULTS
```

```
###############################################################

write.csv(
  rfe_results,
  "rf_rfe_single_split_results.csv",
  row.names = FALSE
)

write.csv(
  all_cv_summary,
  "rf_rfe_all_gene_sets_5fold_cv_summary.csv",
  row.names = FALSE
)

write.csv(
  all_cv_folds,
  "rf_rfe_all_gene_sets_5fold_cv_folds.csv",
  row.names = FALSE
)

write.csv(
  all_cv_predictions,
  "rf_rfe_all_gene_sets_5fold_cv_predictions.csv",
  row.names = FALSE
)

best_rmse_genes <- gene_list[[best_rmse_model$Model]]
best_r2_genes <- gene_list[[best_r2_model$Model]]

write.csv(
  data.frame(gene = best_rmse_genes),
  "rf_best_genes_by_cv_rmse.csv",
  row.names = FALSE
)

write.csv(
  data.frame(gene = best_r2_genes),
  "rf_best_genes_by_cv_r2.csv",
  row.names = FALSE
)

###############################################################
# PART 8: PLOTS
```

```
################################################################

png(
  "rf_all_rfe_gene_sets_cv_RMSE.png",
  width = 2400,
  height = 1800,
  res = 300
)

plot(
  all_cv_summary$n_genes,
  all_cv_summary$Overall_RMSE,
  type = "b",
  pch = 19,
  xlab = "Number of Genes",
  ylab = "Overall CV RMSE",
  main = "Overall CV RMSE Across RFE Gene Sets"
)

points(
  all_cv_summary$n_genes[best_rmse_row],
  all_cv_summary$Overall_RMSE[best_rmse_row],
  pch = 19,
  cex = 2
)

text(
  all_cv_summary$n_genes[best_rmse_row],
  all_cv_summary$Overall_RMSE[best_rmse_row],
  labels = paste0(
    "Best: ",
    all_cv_summary$n_genes[best_rmse_row],
    " genes"
  ),
  pos = 3
)

dev.off()

png(
  "rf_all_rfe_gene_sets_cv_R2.png",
  width = 2400,
  height = 1800,
```

```
  res = 300
)

plot(
  all_cv_summary$n_genes,
  all_cv_summary$Overall_R2,
  type = "b",
  pch = 19,
  xlab = "Number of Genes",
  ylab = "Overall CV R²",
  main = "Overall CV R² Across RFE Gene Sets"
)

points(
  all_cv_summary$n_genes[best_r2_row],
  all_cv_summary$Overall_R2[best_r2_row],
  pch = 19,
  cex = 2
)

text(
  all_cv_summary$n_genes[best_r2_row],
  all_cv_summary$Overall_R2[best_r2_row],
  labels = paste0(
    "Best: ",
    all_cv_summary$n_genes[best_r2_row],
    " genes"
  ),
  pos = 3
)

dev.off()

################################################################
```